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Faculty of Engineering Science
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Fourth-Year Engineering Project

PDR

MagViVit: Transformer-based spatiotemporal magnification for vital signs monitoring

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תקציר

ניטור רציף ביחידות לטיפול נמרץ פגים (NICU) הוא חיוני, אך הפרקטיקה הסטנדרטית מסתמכת על חיישנים פיזיים. חיישנים אלו גורמים לנזקי עור כואבים (MARSI), לסטרס נוירו-התפתחותי הפוגע בהתפתחות המוח, וסובלים מאמינות נמוכה עקב רגישות לארטיפקטים תנועתיים. החידוש המוצע הוא פיתוח מודל למידה עמוקה רב-ערוצי (Multi-modal) ללא מגע, המבוסס על עיבוד מקבילי (Parallel Processing) של וידאו IR ו-NIR, לחיזוי רציף ומדויק של טמפרטורת גוף וקצב נשימה. החידוש טמון בארכיטקטורה היברידית דו-ענפית (Dual-Branch CNN-Transformer). מודל זה משתמש ב-NIR לזיהוי אנטומי וניתוח תנועה, וב-IR ו NIR למדידת טמפרטורה באופן עקיף וזיהוי זרימת אוויר, ליצירת מערכת יתירה (Redundant) ואמינה. השיטה כוללת רישום תמונות (IR-NIR), חילוץ תכונות מרחבי (CNN), ומידול זמני (Transformer) לניתוח תלויות ארוכות טווח וזיהוי אנומליות פיזיולוגיות. תוצאות צפויות: מודל Proof-of-Concept המדגים היתכנות לניטור בדיוק קליני, המאפשר זיהוי מוקדם של התדרדרות ומפחית נזק יאטרוגני.

מילות מפתח: טיפול נמרץ פגים (NICU), ניטור ללא מגע, למידה עמוקה, היתוך רב-ערוצי (Multi-modal Fusion), וידאו NIR, וידאו IR, חיזוי מדדים חיוניים, מידול זמני (Transformer), קצב נשימה, טמפרטורת גוף.

Abstract

Continuous monitoring in the Neonatal Intensive Care Unit (NICU) is vital for preterm infants, yet standard practice relies on adhesive sensors. These sensors cause painful skin injuries (MARSI), neurodevelopmental stress linked to brain dysmaturation, and suffer from low reliability due to motion artifacts. The new method is to develop a contactless, multi-modal deep learning model, based on parallel processing of IR and NIR video, for the continuous and accurate prediction of body temperature and respiratory rate. The innovation lies in a hybrid, dual-branch CNN-Transformer architecture. This model leverages NIR for anatomical context and motion analysis, while utilizing NIR and IR for non-direct thermography and airflow detection, creating a robust and redundant system. The method involves image registration (IR-NIR), spatial feature extraction (CNN), and temporal modeling (Transformer) to analyze long-range dependencies and identify physiological anomalies. Expected Results: A Proof-of-Concept (PoC) model demonstrating clinical-grade accuracy, enabling early detection of deterioration (e.g., sepsis) and reducing iatrogenic harm.

Keywords: Neonatal Intensive Care Unit (NICU), Contactless Monitoring, Deep Learning, Multi-modal Fusion, NIR Video, IR Video, Vital Sign Prediction, Respiratory Rate, Body Temperature, Transformer.

Research Proposal

1. Essence of the Product

The essence of this project is the development of a Proof-of-Concept (PoC) software model. This model will be based on a deep learning architecture capable of continuous, non-contact prediction of vital signs for preterm infants within the Neonatal Intensive Care Unit (NICU). The product will be engineered to ingest and process synchronous video streams. During model training, both NIR video and its converted IR representation will be used, while the final deployed system will operate using only NIR video.
The model's primary output will be a continuous, real-time numerical prediction of two critical vital signs: **Body Temperature (T)** and **Respiratory Rate (RR)**

2. Performance Specifications

The primary goal of the PoC model is to achieve the highest possible accuracy in vital sign prediction. The system's output will be rigorously validated against a "Ground Truth" dataset, and its performance will be benchmarked against current state-of-the-art (SOTA) non-contact monitoring methods.

3. The Technology

The proposed technology is a dual-branch, multi-modal architecture that combines Convolutional Neural Networks (CNNs) and Transformers in two parallel branches.

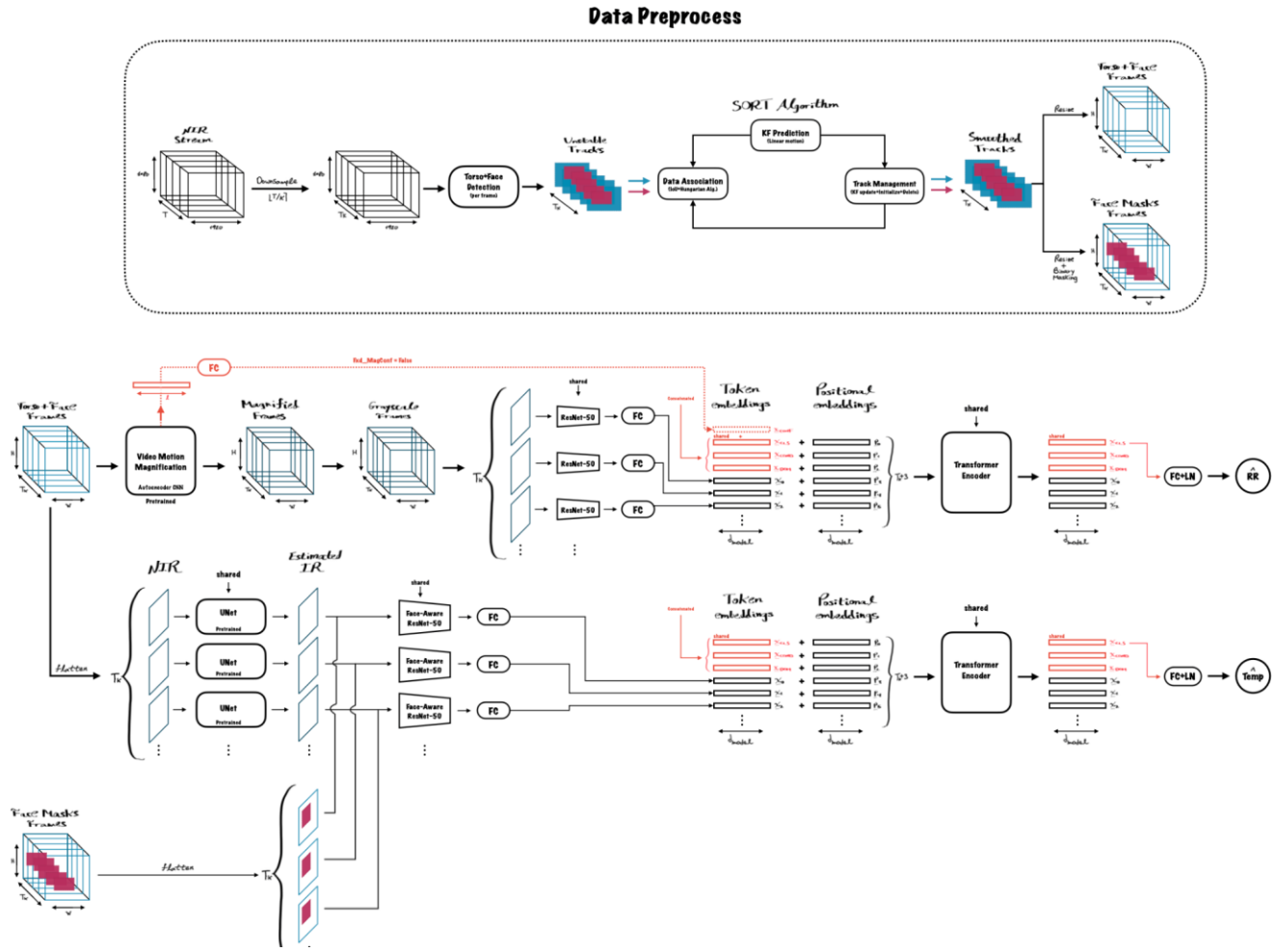
- **CNNs as Spatial Feature Extractors:** Two parallel CNN pipelines will operate on the streams. The RGB-CNN branch (for RR) will use a Pretrained Autoencoder CNN for Video Motion Magnification and a ResNet-50 to extract motion features from Grayscale Frames. The NIR-CNN branch (for Temp) will use a Pretrained UNet to generate Estimated IR maps and a Face-aware ResNet-50 to extract thermal patterns.
- **Transformers as Temporal Modelers:** Two separate Transformer Encoder blocks receive sequences of Token Embeddings (derived from their respective CNNs via an FC layer) along with Positional Embeddings. Their self-attention mechanisms are uniquely suited to model long-range temporal dependencies. This allows them to learn complex patterns across different time scales, from the rapid, short-scale rhythm of respiration (seconds) to the slow,

4. Methodology and Research Phases

The project will be executed in a structured, five-phase pipeline:

1. **Data Acquisition and Calibration:** This foundational phase includes acquiring the clinical dataset of synchronized RGB and NIR videos from the NICU. It involves two critical sub-tasks:
    - **Ground Truth Generation:** This task is crucial for generating the ground truth data and utilizes a hybrid approach: A calibrated RGB camera focused on the monitor, leveraging classical computer vision algorithms (such as edge detection, template matching, or digit classification like MNIST) for initial data extraction. This extracted data will then be synchronized with the video stream.
    - **Image Registration:** A precise calibration process to compute the transformation matrix that spatially aligns the RGB and NIR video streams, likely optimized by maximizing Mutual Information (MI) between the two modalities.
2. **Video Pre-processing:** Preparing the raw video data for the model. This includes using CNNs for object detection (to isolate torso + face regions) and semantic skin segmentation and applying a Video Motion Magnification model (Pretrained Autoencoder CNN) to the NIR stream to amplify subtle respiratory motions. Simultaneously, the NIR stream will be processed by a Pretrained UNet to generate Estimated IR maps.
3. **Data Augmentation:** Implementation of an extensive data augmentation strategy to ensure model robustness. This will include spatial augmentations (rotation, cropping), temporal augmentations, and the addition of synthetic noise and motion artifacts to simulate realistic NICU conditions.
4. **Hybrid Model Development:** This is the core research task of building and training the dual-branch architecture. Pre-trained CNNs (ResNet-50 for NIR stream for both branches) will serve as spatial feature extractors for each frame, and two separate Transformer Encoder architectures will act as temporal modelers to analyze the sequence of these features (as Token Embeddings) over time.
5. **Validation:** The trained model will be rigorously evaluated against a held-out test set to ensure it meets the predefined performance specifications.

System Architecture



The system's workflow begins with a preprocessing pipeline where Torso Face Detection and a Sort Algorithm (using Data Association and KF Prediction) generate Torso + Face Frames and Face Masks Frames.

The preprocessed frames are fed into a neural network split by modality:

- **Respiratory Rate (RR)** Branch: Uses the NIR stream. It applies Video Motion Magnification (via a pretrained Autoencoder CNN) and converts the frames to Greyscale. A shared ResNet-50 extracts spatial features.
- **Temperature (Temp)** Branch: Uses the NIR stream. It applies a pretrained UNet to create estimated IR maps. These maps, along with the Face Masks Frames, are fed into a Face-aware ResNet-50 for feature extraction.

Following spatial extraction, both branches convert their features to Token Embeddings, add Positional Embeddings, and use a shared Transformer Encoder for temporal modeling. Finally, a distinct FC + LayerNorm block in each branch produces the continuous predictions for RR and Temp.

Literature Review

1. Respiratory Rate

The normal respiratory rate (RR) for preterm infants is **30-60 breaths per minute (0.5-1.5 Hz)**. Classical approaches use deterministic algorithms (band-pass filtering and FFT) to identify peak frequencies, or probabilistic models (Bayesian or state-space methods) to fit statistical models to the respiratory waveform [2]. Machine learning is increasingly applied to RR. Deep-learning models (CNNs, transformers, 3D-CNN/LSTM) have been used on neonatal video or thermal data to detect regions of interest and extract RR [1, 3, 4, 6]. These AI models are trained to detect subtle thoracoabdominal (chest and abdomen) movements or changes in skin color from a video feed [1,6]. For example, a transformer+3D-CNN+LSTM model estimated neonatal RR from thermal video with an average of 1.6 bpm error, highlighting AI's potential for robust contactless monitoring[4].

2. Body Temperature

Contactless temperature monitoring is crucial for neonates, as adhesive sensors may damage fragile skin. While infrared thermography is accurate, it is expensive and needs calibration. New deep-learning approaches use RGB and NIR video to infer temperature from visual cues. Clinically, normal infant temperature is **36.5 - 37.5 °C**. Values above 40 °C may indicate fever or prediction error, and readings near 45 °C are non-physiological. Most studies rely on thermal cameras, but a few explored RGB -NIR fusion for indirect estimation. For example, studies have used NIR reflectance to predict skin-temperature variations [7] or achieved strong correlation (R = 0.93) between RGB - NIR predictions and thermography [8]. Yet, no system has directly estimated infant body temperature from RGB + NIR alone - a key gap addressed by this project. Recent work applies transformers for long-range spatiotemporal modeling of heat-related signals. Li et al. [8] introduced a Temporal Vision Transformer (T-ViT) for RGB - NIR fusion, and Nguyen et al. [10] proposed a Spatiotemporal Magnification Transformer (SMT) to amplify subtle emissive changes. These demonstrate the potential of transformer-based multimodal fusion for accurate, contactless thermal prediction.

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